Hyperspectral Image Super-Resolution via Learning an Undercomplete Dictionary and Intra-Algorithmic Postprocessing

Milad Rostami and Ali Asghar Beheshti Shirazi

***Abstract*— This study proposes a novel hyperspectral image super-resolution (HSI-SR) method, whose fundamental requirement to set up is only a *single* low-resolution hyperspectral image (LR-HSI) of the target scene. To fully exploit the inherent features of hyperspectral images (HSIs), including spectral correlation, local sparsity, nonlocal self-similarity, and piecewise smoothness in the spatial domain, we first break down the HSI-SR problem into several identical subcube SR (SC-SR) problems. Then, we design an SC-SR modular model involving two parts: 1) injecting the spatio-spectral information from co-structural subcube patches (SCPs) into each other's sparse representation via learning an undercomplete dictionary and 2) intra-algorithmic postprocessing. The second part, which makes the most of the spatial piecewise smoothness characteristic, consists of two modules, wavelet-based spatial smoothing (SpaS) and adaptive filtering (AF). After merging all the super-resolved SCs and obtaining a high-resolution HSI, we use spectral smoothing (SpeS) to remove the possible remaining distortions in the spectral vector of each pixel. Experiments on the simulated- and real-LR datasets demonstrate the outperformance of our method.**

***Index Terms*— Hyperspectral image (HSI), intra-algorithmic postprocessing, super-resolution, undercomplete dictionary.**

I. Introduction

HIGH spectral resolution in hyperspectral images (HSIs) has led to the utilization of these images in a variety of applications, including agriculture, oil exploration, and environmental monitoring. However, diverse factors, such as sensor noise and limitations in the bandwidth of transmission systems, decrease the spatial resolution of final images taken from devices equipped with hyperspectral sensors [1]. Moreover, due to the tradeoff between spectral and spatial resolutions in imaging systems [2], the issue of low resolution in HSIs cannot be tackled during the capturing phase. Therefore, enhancing the spatial resolution of HSIs is considered a postprocessing problem [3]. It is worth pointing out that in improving the spatial resolution, the spectral consistency of HSIs should be maintained, so HSI super-resolution (HSI-SR) has been propounded as a unique research field to attain these two goals. The SR approaches regarding whether auxiliary images of the target scene are used or not are divided into two general categories: 1) multi-image-based models and 2) single-image-based methods. Techniques based on fusion, the most popular in the first category, intend to achieve a spatially high-resolution HSI through combining the received low spatial resolution HSI with a spatially high-resolution image but spectrally low-resolution image, such as a multispectral image (MSI), three-band MSI (RGB), mosaic RGB, or panchromatic (PAN) image [4], [5], [6], [7], [8], [9]. There are many ways to fuse a low-resolution HSI (LR-HSI) with its corresponding HR-MSI to attain the HR-HSI. Matrix factorization is one of them through which several techniques attack the fusion problem [10], [11], [12]. Tensor-based analysis has recently attracted attention in HSI super-resolution. The coupled sparse tensor factorization (CSTF) [13] is a known method that solved the fusion problem by applying tensor factorization to LR-HSI and HR-MSI in an integrated framework. Dian and Li [14] developed a low-tensor multirank (LTMR) regularizer to solve the fusion issue and exploit HR-HSI prior information. The work [15] made use of the tensor–tensor product (*t-product*) [16] to define a relationship between HR-MSI and HR-HSI and preserve the spatio-spectral information in HSI. Bu et al. [17] designed a novel graph Laplacian-guided coupled tensor decomposition (gLGCTD) model to tackle the fusion HSI-MSI. Deep learning is also a popular approach that has been widely explored recently in fusing MSI and HSI. Dian et al. [18] presented a method incorporating image priors into the fusion model via deep residual neural networks. Dong et al. [19] designed a model-guided deep convolutional network (MoG-DCN) and then mapped a fusion-based HSI-SR algorithm onto their proposed network. In addition, in this category, we can find some models that use the target scene's multiple low-resolution versions with subpixel spatial shifts relative to one another [20].

Although additional images provide more and more priors, getting information from them could be difficult due to the uncertainty of patterns and parameters. Moreover, high-resolution non-HSIs may not always be available. In these cases, we need the second category of SR approaches, i.e., single-image-based methods that aim to enhance the spatial resolution and preserve spectral fidelity when what is available to us is only one low-resolution hyperspectral version of the captured scene. As stated, the fusion-based models exploit the complementary high-resolution versions of the image, e.g., MSI, to produce a spatially and spectrally high-resolution HSI.

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Thus, they are generally more effective than the single-based methods at meeting criteria in the SR process when the image taken has been severely degraded. On the other hand, the single-image-based approaches are more useful if the image resolution is diminished under a moderate distortion scenario and supplementary images of the target scene are unavailable.

From the perspective of problem-solving, the single-image-based methods can be classified into four general groups.

1) Interpolation-based methods. Bilinear and bicubic are well-known ones belonging to this group. They are so fast and straightforward in implementation; however, their outputs generally miss the high-frequency information and have a distortion in the spectral dimension.
2) Regularization-based methods [21], [22], [23]. An HSI-SR is an ill-posed inverse problem; as a result, regularizers based on prior knowledge derived from HSIs' intrinsic features are needed to handle the problem and obtain the best solution. Hence, approaches have emerged that apply the regularizers representing the target image's unique characteristics to the final cost function. Spatial smoothness and spectral correlation are two critical properties of HSIs used as prior knowledge. The work [21] used spatial total variation (STV) minimizing to dictate the spatial smoothness and delimits the HSI-SR problem. Irmak et al. [22] proposed a maximum a posteriori (MAP)-based energy function regularized by the prior information implying the smoothness, unity, and boundaries of endmembers' abundances, and then converted the HSI-SR problem into a quadratic programming one.
3) Sparse-representation-based methods [24], [25], [26], [27]. The existence of similar nonlocal patches and spatially locally low-rank are the other features used in HSI applications. In [24], only the former has been applied, while in [25] and [26], both the items have been used together. To solve the image restoration problem, Zhang et al. [25] proposed a group-based sparse representation in which the dictionary atoms of each group have been learned via singular value decomposition (SVD). In the HSI-SR problem, Li et al. [26] have acted the same as [25] in terms of using the grouping concept and learning the spectral dictionary.
4) Deep-learning-based techniques [28], [29], [30]. Similar to fusion-based methods, the single-image-based algorithms have also extensively used deep learning. Yang et al. [28] proposed a multiscale wavelet 3-D convolutional neural network (MW-3D-CNN) consisting of two subnetworks whose tasks are extracting deep features of LR-HSI and estimating the wavelet coefficients of HR-HSI. In [29], a novel gradient-guided residual dense network (G-RDN) is proposed to produce an HSI-SR algorithm that makes an end-to-end connection between LR-HSI and HR-HSI through which the gradient detail is imported to HR-HSI.

Although single-image-based schemes do not need any high-resolution image of the *target* scene, in terms of their other requisites such as images of different locations, they could be divided into three classes.

1) Models that require low-resolution versions at all the spectral bands to reconstruct a typical scene at one spectral channel, such as Bayesian-based methods [22] and [31].
2) Techniques that need images from other scenes to set up. The methods with training phases mainly lie in this class, like the literature [27], which uses vast training imagery samples to learn a pretrained dictionary. Deep-learning-based methods [28], [29], [30] are also among others that use many LR–HR pairs of imagery samples to train neural networks. Like Yang et al. [28], these techniques had better be called the single-*frame*-based rather than single-image-based.
3) Algorithms that require only original LR-HSI and can enhance the spatial structure of a captured HSI at each band independently of other spectral channels. The bicubic method, SSG [26], and our method belong to this class. Although the approaches in the third class may suffer from sensitivity to parameters' tuning, they are superior to methods in the other two categories in terms of simplicity in setting up and not requiring a training phase (consequently not requiring other scenes).

In this study, we propose a novel single-image-based super-resolution method that is categorized in the third class. Based on the inherent characteristics of HR-HSIs, we first break down the problem of low-resolution HSI super-resolution into identical subproblems of low-resolution subcubes super-resolution (LR-SCs-SR). To solve these subproblems, which could be resolved parallelly, we have designed a modular model consisting of two main parts: 1) injecting spatio-spectral information from co-structural subcube patches (SCPs), which are clustered by a rotation-invariant criterion, into each other via a one-stage dictionary learning (OSDL) algorithm and 2) intra-algorithmic postprocessing. The first part exploits three prior information of HR-HSIs simultaneously: nonlocal self-similarity, spatial local sparsity, and the fact that pixels could be spectrally represented in a low-dimensional subspace. The second part consists of two modules, wavelet-based spatial smoothing (SpaS) and adaptive filtering (AF), which make the most of two other features of HR-HSIs in the spatial domain: being low-pass in a typical frequency transformation and piecewise smoothness. The model's parts and the corresponding modules are embedded in an iterative framework by forming a simple objective function and using the alternating direction method of multipliers (ADMM) to solve it. Merging all the super-resolved SCs, we obtain an HR-HSI. Finally, to eliminate the possible remaining distortions from the spectral dimension of this HR-HSI, we use wavelet-based spectral smoothing (SpeS) on the spectral vector of each pixel. Fig. 1 depicts the diagram of the proposed method.

The remainder of this article is organized as follows. Section II explains the proposed method step by step; Section III provides quantitative and visual simulation results and comparisons with the state-of-the-art methods. Section IV presents the conclusion, and finally, three appendices describing the concepts we have used in this article are given at the end.

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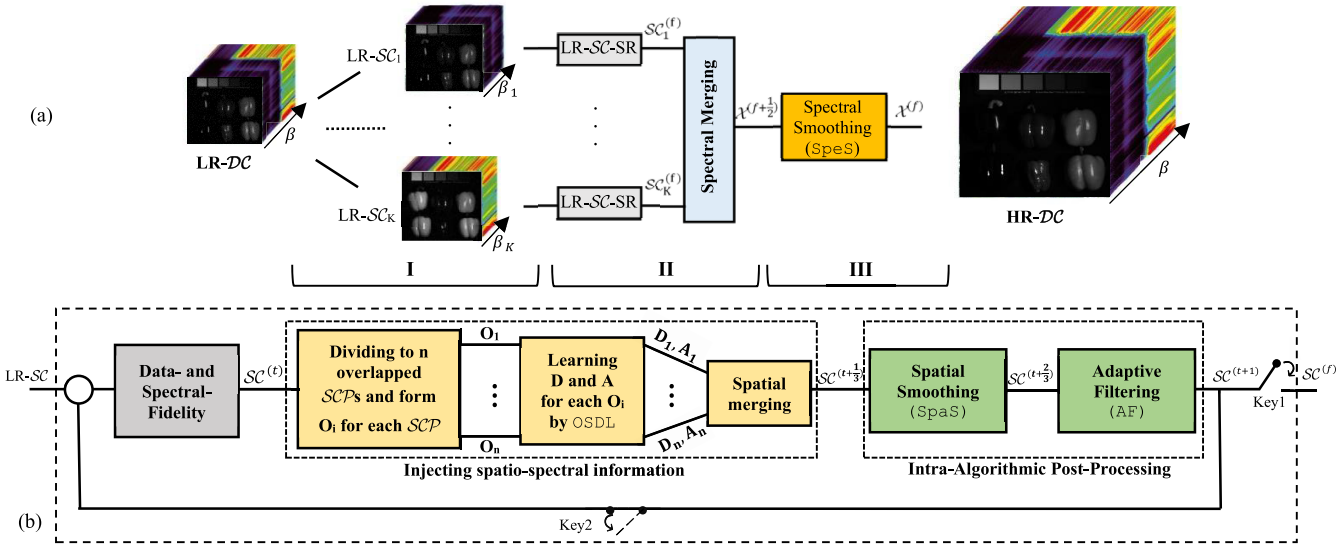


Fig. 1. (a) Diagram of LR-DC-SR by the LR-SC-SR algorithm and SpeS. (b) Diagram of the LR-SC-SR model; when Key 2 is OFF, Key 1 will be ON.

II. Proposed Approach

A. Problem Modeling

HSI is generally considered a data cube (DC) consisting of a significant number of channels that present the same scene in different spectra. So, the HR and LR versions of an HSI can be represented as the third-order tensors $\mathcal{X} \in \mathbb{R}^{m_x \times n_x \times \beta}$ and $\mathcal{Y} \in \mathbb{R}^{m_Y \times n_Y \times \beta}$, respectively, where β, $m_x n_x$ and $m_Y n_Y$ are the number of spectral bands and the number of pixels in each spectral channel of $\mathcal{X}$ and $\mathcal{Y}$, respectively. A DC results from merging subcubes (SCs). The tensor $\mathrm{SC}_k \in \mathbb{R}^{m_x \times n_x \times \beta_k}$, $(k = 1, 2, \ldots, K)$, represents a typical SC, where β_k is the number of its bands. A typical SC_k, in turn, consists of SCPs, each of which is denoted by the tensor $\mathrm{SCP}_k \in \mathbb{R}^{m_p \times n_p \times \beta_k}$, with β_k bands and a spatial pixel size of $m_p n_p$.

We can map the M-dimensional tensor $\mathcal{H}$ onto its matrix form $\mathbf{H}_{(n)}$ and vice versa via functions $\mathtt{Ten2Mat}^{(n)}(\mathcal{H})$ and $\mathtt{Mat2Ten}^{(n)}(\mathbf{H}_{(n)})$, as follows:

$$\mathcal{H} \in \mathbb{R}^{d_1 \times, d_2 \times, \ldots, d_n, \ldots, \times d_M} \underset{\mathtt{Mat2Ten}^{(n)}}{\overset{\mathtt{Ten2Mat}^{(n)}}{\rightleftarrows}} \mathbf{H}_{(n)} \in \mathbb{R}^{d_n \times (d_1, d_2, \ldots, d_{n-1} d_{n+1}, \ldots, d_M)} \tag{1}$$

where $\mathbf{H}_{(n)}$ is called the n-mode matricization of the tensor $\mathcal{H}$. It is clear that

$$\mathcal{H}(:) = \mathbf{H}_{(n)}^T(:) = \mathbf{H}^T(:), \quad \text{if } n = M \tag{2}$$

where "T" denotes the transpose of tensor, matrix, or vector; moreover, $\mathcal{H}(:)$ and $\mathbf{H}_{(n)}^T(:)$ represent the vector form of $\mathcal{H}$ and $\mathbf{H}_{(n)}^T$, respectively. This article presents the n-mode matricization of the tensor $\mathcal{H} \in \mathbb{R}^{d_1 \times, d_2 \times, \ldots, d_n, \ldots, \times d_M}$ via $\mathbf{H}$ when $n = M$.

If the lth spectral band of the HR-HSI $\mathcal{X}(:, :, l)$ and LR-HSI $\mathcal{Y}(:, :, l)$, $l \in [\beta]$, is denoted by $\mathbf{X}^l \in \mathbb{R}^{m_x \times n_x}$ and $\mathbf{Y}^l \in \mathbb{R}^{m_Y \times n_Y}$, respectively, we will have

$$\begin{aligned} \mathbf{X} &= \left[\mathbf{X}^1(:)\middle|, \ldots, \middle|\mathbf{X}^1(:)\middle|, \ldots, |\mathbf{X}^\beta(:)\right]^T \in \mathbb{R}^{\beta \times (m_x n_x)} \\ \mathbf{Y} &= \left[\mathbf{Y}^1(:)\middle|, \ldots, \middle|\mathbf{Y}^l(:)\middle|, \ldots, |\mathbf{Y}^\beta(:)\right]^T \in \mathbb{R}^{\beta \times (m_Y n_Y)}. \end{aligned} \tag{3}$$

The tube $\mathcal{X}(i, j, \Gamma) = \mathcal{X}_{i,j}^{\Gamma}$, where the pair (i, j) is constant and Γ is variable, denotes the intensity values of a pixel at coordinates (i, j) as a function of spectrum number. We name its vector form as the spectral reflectance function (SRF) of the pixel (i, j), which is defined as

$$\mathrm{SRF}_{i,j}(\Gamma) = \mathcal{X}_{i,j}^{\Gamma}(:). \tag{4}$$

According to the HSI acquisition linear model [33], the relationship between $\mathbf{X}$ and $\mathbf{Y}$ is considered as follows:

$$\mathbf{Y} = \mathbf{XBS} + \mathbf{N}, \quad \mathbf{Z} = \mathbf{BS} \tag{5}$$

$$\mathbf{Y} = \mathbf{XZ} + \mathbf{N} \tag{6}$$

where $\mathbf{B} \in \mathbb{R}^{m_x n_x \times m_x n_x}$, $\mathbf{S} \in \mathbb{R}^{m_x n_x \times m_Y n_Y}$, $\mathbf{N} \in \mathbb{R}^{\beta \times m_Y n_Y}$, and $\mathbf{Z} \in \mathbb{R}^{m_x n_x \times m_Y n_Y}$ are the blurring kernel, the down-sampler operator, the additive noise, and the overall degradation operator, respectively. $\mathbf{N}$ is generally assumed to be an additive white Gaussian noise (AWGN) signal.

B. Inherent HR-HSIs Features and Breaking the SR Problem

This section expresses the features that generally exist in an HR-HSI (**F1–F5**), based on which we will design our HSI-SR method.

1) **F1:** The intensity values of a pixel in different spectra (especially in adjacent spectra) evince high correlations [34].
2) **F2:** There are patches with similar structures throughout the image [24]; this property is referred to as nonlocal self-similarity.
3) **F3:** In natural images, the pixels of a typical patch could be represented in a low-dimensional space [35]; this item is known as local sparsity.
4) **F4:** $\mathbf{X}^l = \mathcal{X}(:, :, l)$, $l \in [\beta]$, the lth band of the HR$-DC$, is a piecewise smooth manifold with a limited number of points with abrupt jumps created by edges and high-frequency parts [36].

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5) **F5:** In a typical frequency transformation, such as Fourier or wavelet transforms, most energy, and consequently most information of a natural image, is centralized in the lower frequency parts of that transform [37].

An HSI is captured at a continuum of wavelengths. Moreover, according to **F1**, the signatures of a pixel in adjacent spectra are highly correlated. Accordingly, it can be generally concluded that sudden changes are not observed in the intensity value of a pixel in the spectral dimension, that is, $\mathrm{SRF}_{i,j}(\Gamma)$ can be considered a 1-D smooth function. A smooth function is definitely piecewise smooth; thus, to speed up the overall super-resolution process, instead of facing an LR-DC-SR problem, we can break it down into LR-SCs-SR subproblems, which could be solved independently of each other. To ensure that merging all the super-resolved SCs produce a DC with smooth SRFs, the LR-DC is broken down such that the adjacent LR-SCs overlap in one or two bands.

These K overlapped LR-SCs in matrix form are as follows:

$$\begin{aligned}\mathbf{LR_SC}_1 &= \mathbf{Y}_1 = \left[\mathbf{Y}^1(:)|, \ldots, |\mathbf{Y}^{\beta_1}(:)\right]^T \in \mathbb{R}^{\beta_1 \times m_Y n_Y}\\ \mathbf{LR_SC}_2 &= \mathbf{Y}_2 = \left[\mathbf{Y}^{\beta_1}(:)|, \ldots, |\mathbf{Y}^{\beta_2+\beta_1-1}(:)\right]^T \in \mathbb{R}^{\beta_2 \times m_Y n_Y}\\ \mathbf{LR_SC}_K &= \mathbf{Y}_K = \left[\mathbf{Y}^{\beta-\beta_K+1}(:)|, \ldots, |\mathbf{Y}^{\beta}(:)\right]^T \in \mathbb{R}^{\beta_K \times m_Y n_Y}.\end{aligned} \tag{7}$$

As the LR-DC-SR problem is broken into several identical LR-SCs-SR subproblems, we now focus on describing the LR-SC-SR model. As depicted in Fig. 1(b), this model consists of two parts: injecting spatio-spectral information and intra-algorithmic postprocessing. We explain them in the following.

C. Injecting Spatio-Spectral Information

According to **F1**, we can represent a pixel in a low-dimensional spectral subspace. In addition, we can interpret from **F2** that the information lost in a particular patch could be found in the patches co-structural to it. For example, some atoms in a patch might be removed by down-sampling, blurring, or noise, while still remaining in patches similar to it. Also, based on **F3**, a patch is sparse in the subspace defined by the patch's constituent atoms. To take advantage of these features in an integrated manner, we first divide the kth primary high-resolution $\mathrm{SC}_k \in \mathbb{R}^{m_x \times n_x \times \beta_k}$ (see (22) at the end of this section) into n SCPs with overlapped pixels (SCP_1^k, $\mathrm{SCP}_2^k, \ldots, \mathrm{SCP}_n^k \in \mathbb{R}^{m_p \times n_p \times \beta_k}$). Then for each SCP_i^k, $(i = 1, 2, \ldots, n)$, we form the array $\mathcal{G}_i^k$ consisting of SCP_i^k itself and $C-1$ SCPs, which are rotationally similar to it. To cluster SCPs being rotationally co-structural to SCP_i^k, we consider a 3-D window of size $L \times L \times \beta_k$ with the centrality of SCP_i^k. Then, we use a criterion based on fast Fourier transform (FFT) that is able to recognize patches within the window, which are similar to SCP_i^k or to its rotated versions with an arbitrary angle. This rotation-invariant criterion has been described in detail in Appendix A.

The array $\mathcal{G}_i^k$, called the grouping array of SCP_i^k, has the form as follows:

$$\mathcal{G}_i^k = \left[\mathrm{SCP}_i^k \middle|^{\circlearrowleft 1}\mathrm{SCP}_i^k\middle|, \ldots, \middle|^{\circlearrowleft C-1}\mathrm{SCP}_i^k\right] \in \mathbb{R}^{m_p \times n_p \times \beta_k \times C} \tag{8}$$

where ${}^{\circlearrowleft j}\mathrm{SCP}_i^k$ represents the jth SCP that is rotationally similar to SCP_i^k. Moreover, we define the matrix $\mathbf{O}_i^k$ as follows:

$$\left(\mathtt{Ten2Mat}^{(4)}\left(\mathcal{G}_i^k\right)\right)^T = \left(\mathbf{G}_i^k\right)^T = \mathbf{O}_i^k \in \mathbb{R}^{m_p n_p \beta_k \times C}. \tag{9}$$

We call $\mathbf{O}_i^k$ the observation matrix of SCP_i^k because the other SCPs similar to SCP_i^k can be considered the other observed versions of it. It is obvious that the mathematical interpretation of the observation matrix is

$$\mathbf{O}_i^k = \left[\mathrm{SCP}_i^k(:)\middle|^{\circlearrowleft 1}\mathrm{SCP}_i^k(:)\middle|, \ldots, \middle|^{\circlearrowleft C-1}\mathrm{SCP}_i^k(:)\right]. \tag{10}$$

Now we can extract the dictionary atoms and sparse coefficients of SCP_i^k from its observation array (it will be stated why we chose this arrangement for $\mathbf{O}_i^k$, in the following). Using an array gathering co-structural patches is widespread in many sparse-based HSI-SR approaches [14], [26]. These methods generally use the SVD to learn the dictionary atoms and then *separately* estimate the sparse coefficients based on a cost function regularized by prior information. SVD obtains the following factorization for O_i^k:

$$O_i^k = U_i \Sigma_i V_i{}^T = \sum \alpha_i \left(u_i v_i^T\right). \tag{11}$$

From (11), we can observe that SVD represents O_i^k as a linear combination of the atoms $u_i v_i^T$, such that singular values α_is are the coefficients of this representation. Therefore, the *separate* estimation of coefficients could be considered a sparsifying process on α_is, which causes the elimination of some atoms playing a role in the representation of SCP_i^k. Especially when O_i^k is not low-rank enough; consequently, singular values are not sparse enough. For example, in the case that SCPs gathered in O_i^k do not have a resemblance enough relative to each other; a sparsifying process has more influence in eliminating atoms of the representation of SCP_i^k.

In this study, to improve the accuracy of the sparse representation, inspired by Sadeghi et al. [38], we use the OSDL algorithm, which learns dictionary atoms and sparse coefficients *simultaneously*. The dictionary learning algorithm we have used to extract the sparse representation of $\mathbf{O}_i^k$ is given in **Algorithm 1**, in which $d_{\mathbf{O}_i^k}^{\tau}$ is the τth column of the dictionary matrix $\mathbf{D}_{\mathbf{O}_i^k}$ and $a_{\mathbf{O}_i^k}^{[\tau]}$ is the τth row of coefficients' matrix $\mathbf{A}_{\mathbf{O}_i^k}$. Moreover, "num" and $\mathrm{Softhr}\left[., \lambda\right]$ denote the number of atoms and the soft-threshold function [39] with the threshold value λ, respectively.

The main point that should be stated is that OSDL generally is an overcomplete dictionary learning algorithm, i.e., the number of atoms is more than the number of dimensions of the signal to be represented sparsely. So, at first glance, it seems that "num" should be more than $\beta_k m_p n_p$; it results in high dimensions of $\mathbf{D}_{\mathbf{O}_i^k}$ and increases the computational burden. However, we have used OSDL to learn an undercomplete dictionary based on the following argument. On one hand, SCP(:) is indeed the vectorized version of a *particular* patch in β_k spectra; obviously, the *percentage* of atoms spatially composing a patch is constant in different spectra; thus, the dimensions of the spatial subspace in which a particular patch is represented are equal in all the spectral channels. On the other hand, the dimensions of the spectral subspace where

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Algorithm 1: OSDL

Task: Learning dictionary matrix $\mathbf{D}_{\mathbf{O}_i^k} \in \mathbb{R}^{\beta_k m_p n_p \times num}$ and coefficients matrix $\mathbf{A}_{\mathbf{O}_i^k} \in \mathbb{R}^{\text{num} \times C}$ for $\mathbf{O}_i^k$
Initialization $\mathbf{D}_{\mathbf{O}_i^k} = \mathbf{D}_{\mathbf{O}_i^k}{}^{(0)}$, $\mathbf{A}_{\mathbf{O}_i^k} = \mathbf{A}_{\mathbf{O}_i^k}{}^{(0)}$, num, R_m, $\mathbf{E} = \mathbf{O}_i^k - \mathbf{D}_{\mathbf{O}_i^k}\mathbf{A}_{\mathbf{O}_i^k}$, λ
Repeat:
  for $mm = 1 : R_m$
    for $\tau = 1 : \text{num}$
      $\mathbf{E}_\tau = \mathbf{E} + d^\tau_{\mathbf{O}_i^k} a^{[\tau]}_{\mathbf{O}_i^k}$
      $\left(a^{[\tau]}_{\mathbf{O}_i^k}\right)^T = \text{Softhr}[\mathbf{E}_\tau^T d^\tau_{\mathbf{O}_i^k}, \lambda]$
      $d^\tau_{\mathbf{O}_i^k} = \mathbf{E}_\tau \left(a^{[\tau]}_{\mathbf{O}_i^k}\right)^T$
      $d^\tau_{\mathbf{O}_i^k} = d^\tau_{\mathbf{O}_i^k} / \left\| d^\tau_{\mathbf{O}_i^k} \right\|_2$
      $\mathbf{E} = \mathbf{E}_\tau - d^\tau_{\mathbf{O}_i^k} a^{[\tau]}_{\mathbf{O}_i^k}$
    end
  end
Until convergence

each pixel lives generally are fewer than a patch size in one spectrum. Therefore, it is sufficient that the number of atoms is more than an SCP size in the spatial domain, i.e., $\text{num} > m_p m_p$. Indeed, the reason why we have chosen the arrangement (10) for $\mathbf{O}_i^k$ is that it enables us to use OSDL in learning an undercomplete dictionary without deviating from its best performance. As the sparse representation of the observations' matrix is $\mathbf{O}_i^k \cong \mathbf{D}_{\mathbf{O}_i^k}\mathbf{A}_{\mathbf{O}_i^k}$, the sparse representation of SCP_i^k is achieved from (10) as follows:

$$\hat{\text{SCP}}_i^k(:) = \mathbf{O}_i^k(:, 1) = \mathbf{D}_{\mathbf{O}_i^k}\mathbf{A}_{\mathbf{O}_i^k}(:, 1) = \mathbf{D}_{\mathbf{O}_i^k} a_{\mathbf{O}_i^k} 1 \tag{12}$$

where $a^1_{\mathbf{O}_i^k}$ is the first column of $\mathbf{A}_{\mathbf{O}_i^k}$. So, the new version of SC_k would be obtained by merging the n $\widehat{\text{SCP}_i^k}$

$$\text{SC}_k = \sum_{i=1}^{n} \psi_i\left(\widehat{\text{SCP}_i^k}\right) \cdot \Big/ \sum_{i=1}^{n} \psi_i\left(1_{m_p \times n_p \times \beta_k}\right) \tag{13}$$

where $\boldsymbol{1}_{m_p \times n_p \times \beta_k}$ is a third-order all-one tensor, and $\psi_i(\cdot)$ puts the $\widehat{\text{SCP}_i^k}$ and $\boldsymbol{1}_{m_p \times n_p \times \beta_k}$ in the ith parts of an all-zero tensor with dimensions $m_x \times n_x \times \beta_k$. Following (12) and (13), SC_k and its matrix form can be considered a function of the dictionaries and sparse coefficients of all SCP_i^ks, $(i = 1, 2, \ldots, n)$. We denote this function by $\boldsymbol{\Phi}$

$$\mathbf{SC}_k = \boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k) \tag{14}$$

where $\boldsymbol{\Delta}_k$ and $\boldsymbol{\Lambda}_k$ are the concatenation of all $\mathbf{D}_{\mathbf{O}_i^k}$s and all $a^1_{\mathbf{O}_i^k}$s, $(i = 1, 2, \ldots, n)$, respectively, as follows:

$$\boldsymbol{\Delta}_k = \left[\mathbf{D}_{\mathbf{O}_1^k} \middle| \mathbf{D}_{\mathbf{O}_2^k} \middle| \cdots \middle| \mathbf{D}_{\mathbf{O}_n^k}\right]$$
$$\boldsymbol{\Lambda}_k = \left[a^1_{\mathbf{O}_1^k} \middle| a^1_{\mathbf{O}_2^k} \middle| \cdots \middle| a^1_{\mathbf{O}_n^k}\right]. \tag{15}$$

Now, we design the subcube SR (SC-SR) loop by forming the following simple cost function:

$$\widehat{\mathbf{SC}}_k = \min_{\mathbf{SC}_k} \frac{1}{2}\|\mathbf{SC}_k\mathbf{Z} - \mathbf{Y}_k\|_F^2 + \theta\|\mathbf{TSC}_k\|_{1,1} \tag{16}$$

where $\|\cdot\|_{1,1}$ stands for the so-called entrywise matrix l_1 norm. Moreover, $\mathbf{T}$ denotes the spectral total variation operator. The mathematical definitions of the matrix l_1 norm and spectral total variation operator are given in Appendix B.

Taking (14) into account, we can convert the cost function of (16) as follows:

$$\Phi(\Delta_k, \Lambda_k) = \min_{\boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)} \frac{1}{2}\|\boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)\mathbf{Z} - \mathbf{Y}_k\|_F^2 + \theta\|\mathbf{T}\Phi(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)\|_{1,1}. \tag{17}$$

As we aim to design an iterative scheme, we use the framework of ADMM [40]. Accordingly, we rewrite (17) by introducing the auxiliary variable $\mathbf{U}_k$ as

$$\Phi(\Delta_k, \Lambda_k) = \min_{\boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)} \frac{1}{2}\|\mathbf{U}_k\mathbf{Z} - \mathbf{Y}_k\|_{\mathbf{F}}^2 + \theta\|\mathbf{T}\Phi(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)\|_{1,1} \quad \text{s.t } \boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k) = \mathbf{U}_k. \tag{18}$$

The augmented Lagrangian function of (18) is as follows:

$$\mathcal{L}_\mu(\boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k), \mathbf{U}_k, \mathbf{V}) = \frac{1}{2}\|\mathbf{U}_k\mathbf{Z} - \mathbf{Y}_k\|_F^2 + \theta\|\mathbf{T}\Phi(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)\|_{1,1} + \mu\left\|\mathbf{U}_k - \boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k) + \frac{\mathbf{V}}{2\mu}\right\|_F^2 \tag{19}$$

where $\mathbf{V}$ is the Lagrangian multiplier matrix, and $\mu > 0$ is the penalty parameter. The alternative iterations to solve (19) are as follows:

$$\mathbf{U}_k^{(t+1)} = \text{argmin}_{\mathbf{U}_k} \mathcal{L}_\mu\left(\boldsymbol{\Phi}^{(t)}\left(\boldsymbol{\Delta}_k{}^{(t)}, \boldsymbol{\Lambda}_k{}^{(t)}\right), \mathbf{U}_k, \mathbf{V}^{(t)}\right) \tag{20}$$

$$\boldsymbol{\Phi}^{\left(t+\frac{1}{2}\right)}\left(\boldsymbol{\Delta}_k{}^{(t)}, \boldsymbol{\Lambda}_k{}^{(t)}\right) = \text{argmin}_{\Phi(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)} \mathcal{L}_\mu\left(\boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k), \mathbf{U}_k{}^{(t+1)}, \mathbf{V}^{(t)}\right). \tag{21}$$

Since the inner update of $\boldsymbol{\Phi}(\boldsymbol{\Delta}_k, \boldsymbol{\Lambda}_k)$ at the beginning of one iteration in the SC-SR loop is generated at this stage, we use the superscript $(t + 1/2)$ to denote its update in this phase.

The details of solving the two above optimization problems (20) and (21) are presented in Appendix B.

Note that the Lagrangian multiplier $\mathbf{V}$ would not be updated at this stage.

We consider the primary HR-SC at the beginning of each iteration as follows:

$$\mathbf{SC}_k^{(t)} = \boldsymbol{\Phi}^{\left(t+\frac{1}{2}\right)}\left(\boldsymbol{\Delta}_k{}^{(t)}, \boldsymbol{\Lambda}_k{}^{(t)}\right). \tag{22}$$

Thus, what to be divided into n SCP_i^ks is the tensor $\mathbf{SC}_\mathbf{k}^{(t)}$. Forming the $\mathbf{O}_i^k$s and implementation OSDL algorithm on them, we can achieve the updated $\Delta_k{}^{(t+1)}$, and $\Lambda_k{}^{(t+1)}$. Consequently, we can yield the next inner updated version SC_k, whose superscript is denoted by $(t + 1/3)$:

$$\mathbf{SC}_k^{(t+1/3)} = \boldsymbol{\Phi}^{\left(t+\frac{1}{2}\right)}\left(\boldsymbol{\Delta}_k{}^{(t+1)}, \boldsymbol{\Lambda}_k{}^{(t+1)}\right). \tag{23}$$

D. Intra-Algorithmic Postprocessing and SC-SR Algorithm

So far, we have not exerted any **F4-** and **F5**-based regularizers in the cost function. Instead, we introduce the concept of *intra-algorithmic postprocessing* through which $\text{SC}_k^{(t+1/3)}$ is processed by two modules before proceeding to the next iteration of the loop. These two modules are SpaS and AF, which we address in the following.

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1) Spatial Smoothing: $\mathrm{SC}_k^{(t+1/3)}$ has been obtained without applying **F4** and **F5**, so it contains noise and artifacts with the energy distributed at all the components of a typical frequency transformation. Recalling **F5,** we can consider this distribution in two parts: the part outside the frequency range of the valuable information of the image and the other inside the image frequency range. To eliminate the first part, we used the 2-D wavelet packet-based smoothing such that, at first, $N1$-scale decomposition would be performed on each spectral band of the $\mathrm{SC}_k^{(t+1/3)}$, and then each smoothed band of $\mathrm{SC}_k^{(t+2/3)}$ is obtained by zeroing the $N1$th level details' coefficients in the three directions (horizontal, vertical, and diagonal) and computing the wavelet transform's inverse. The wavelet function should be type-matched to the natural images and fine enough to preserve the edges and the high-frequency information.

2) Adaptive Filtering: We use the AF conception to apply **F4**, attenuate the noise and artifacts distributed inside the frequency range of $\mathrm{SC}_k^{(t+2/3)}$, and finally, to attain $\mathrm{SC}_k^{(t+1)}$. The adaptive filter selected in this article is the low-pass pixelwise Wiener filter [41]. One of the reasons for selecting this filter is its excellent performance in removing noise with constant power, like AWGN, which is generally used in image noise modeling. As shown in Appendix C, the other reason is that this filter locally estimates the input parameters (such as the noise power) needed to achieve the output from the degraded image itself. It means that the only requisites of this filter are the distorted image and the size of the neighborhood in which the required parameters are computed.

We explain our *motivation* for applying AF as follows. In many previous works, to bound the SR problem and weaken artifacts in the spatial domain, the regularizers derived from the spatial smoothness were exerted, such as the STV [21] or the energy difference between a typical pixel and its neighbors in abundances domain [22]. Following **F4**, in practice, a natural image is spatially *piecewise* smooth, so it has regions with different local variances, regions with small variances (homogenous regions), and regions with significant variances (the regions containing the edges and high-frequency information). As the STV is a sparsifying operator, it preserves the edges but causes over-smoothing in homogeneous regions. On the other hand, the smoothness regularizer in [22] is based on the Markov random field (MRF) model, whose assumption is that the spatial characteristics have no abrupt variations in a neighborhood [42]. Although this regularizer preserves the valuable information in homogeneous regions, it results in high-frequency information loss, and sharpness in the final super-resolved image will not be maintained well. In this study, instead of using such terms in the final *cost function,* the Wiener filter, which acts with attention to the local variances of the image, is embedded as the SC-SR *algorithm*'s module and applied to each spectral tube (spectral vector of each pixel) of $\mathrm{SC}_k^{(t+2/3)}$. It leads to the preservation of sharpness and prevents homogeneous regions from over-smoothing. The neighborhood size (NS) determines the power of this filter in noise and artifact removal, and like the balancing factors in regularizers, it is obtained heuristically. Since the upper and lower limits of the NS are odd natural numbers ($\mathbb{N}$), finding their best values is by far more straightforward than finding the balancing factor's best value in the domain of the real numbers ($\mathbb{R}$). The larger the NS, the higher the filter's power.

Now by obtaining $\mathbf{\Phi}^{(t+1)}\left({\mathbf{\Delta}_k}^{(t+1)}, {\mathbf{\Lambda}_k}^{(t+1)}\right) = \mathrm{SC}_k^{(t+1)}$, the matrix $\mathbf{V}$ would be updated as follows:

$$\mathbf{V}^{(t+1)} = \mathbf{V}^{(t)} + \mu\left(\mathbf{U}_k^{(t+1)} - \mathbf{\Phi}^{(t+1)}\left({\mathbf{\Delta}_k}^{(t+1)}, {\mathbf{\Lambda}_k}^{(t+1)}\right)\right). \quad (24)$$

Updating $\mathbf{V}$ at this stage underlines that SpaS and Wiener are the modules of the SC-SR algorithm and must not be considered postprocessing merely. That is why we call their combination *intra-algorithmic* postprocessing. The summarization of the super-resolution process for an SC is shown in Algorithm 2.

E. SpeS and Obtaining Final HR-DC

By accomplishing the super-resolution of all the LR-SCs and achieving $\mathrm{SC}_k^{(f)} \in \mathbb{R}^{m_x \times n_x \times \beta_k}$, $(k = 1, 2, \ldots, K)$, the HR-DC $\mathcal{X}^{(f+1/2)}$ is obtained by merging them and averaging over the overlapped bands as follows (see Fig. 1(a)-II):

$$\mathcal{X}^{(f+1/2)} = \sum_{k=1}^{K} \Psi_k\left(\mathrm{SC}_k{}^{(f)}\right) \cdot \Big/ \sum_{k=1}^{K} \Psi_k\left(1_{m_x \times n_x \times \beta_k}\right) \quad (25)$$

where $1_{m_x \times n_x \times \beta_k}$ is a third-order all-one tensor; the operator Ψ_k puts $\mathrm{SC}_k^{(f)}$ and $1_{m_x \times n_x \times \beta_k}$ in the kth parts of a third-order all-zero tensor with dimensions $m_x \times \beta$, and ./ is the elementwise division operator.

Since smoothing and filtering in Algorithm 2 have been carried out in the spatial dimensions, it can be claimed that the SRFs may still have distortions. To eliminate the possible distortions in the SRF of each pixel, we applied 1-D wavelet-based smoothing in spectral dimension and pixel-by-pixel (on the SRF of each pixel), which we call SpeS. As shown in Fig. 1, in Algorithm 2, when the iterations reach $\mathrm{iter}_{\max}$, key_2 is opened and key_1 is closed.

Note that in `SpeS`, we generally use 1-D wavelet-based smoothing, which zeros the detail coefficients in the $N2$th level of the wavelet decomposition and then computes the modified transform's inverse.

However, in some cases, particularly when the HSI has many spectral bands, like a remote sensing HSI, we use the traditional denoising based on the interval dependent [43], [44]. This method is implemented straightforwardly by the built-in function `cmddenoise(sig,wname,level)`, which is contained in the wavelet toolbox of MATLAB. Moreover, in some other cases, `SpeS` may not yield the desired improvement, and it is construed as the decomposition into the zero level ($N2 = 0$), i.e., no `SpeS` is applied.

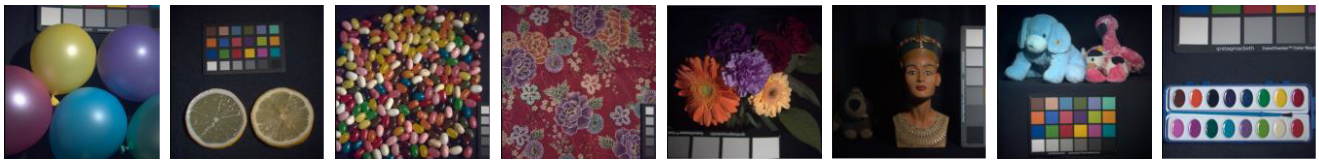
III. Simulations

A. Datasets, Distortion Scenario, and Quality Metrics

To evaluate the proposed method, we have run it on LR-HSIs simulated from two datasets.

Dataset A is Columbia computer vision laboratory (CAVE) [45], containing 32 indoor 512 × 512-pixeled HSIs captured at 31 spectral channels over the range of 400–700 nm

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 2. Some test images from dataset A [45]. For a better view, the HR-RGB versions of these images are shown. (a) Balloons. (b) Lemon slices. (c) Jelly beans. (d) Cloth. (e) Flowers. (f) Egyptian statue. (g) Stuffed toys. (h) Paints.

Algorithm 2 : SC_k Super-Resolution

Input: LR $\mathcal{SC}_k$, degradation operator $\mathbf{Z}$, β_k, $m_x n_x$ n, $m_p n_p$, C, μ,θ, N1, wavelet-type (WT), neighborhood size (NS), $iter_{max}$.

Initialization: $\mathbf{V}^{(0)}=0$, $\mathbf{U}_k^{(0)}=0$, $\mathbf{\Phi}^{(0)}=0$

for $t = 0 : iter_{max}$ do
  Update $\mathbf{U}_k^{(t+1)}$ via (20)
  Update $\mathbf{SC}_k^{(t)}$ via (21), and (22)
  $\mathcal{SC}_k^{(t)} = \texttt{Mat2Ten}^{(3)}\left(\mathbf{SC}_k^{(t)}\right)$
  Divide $\mathcal{SC}_k^{(t)}$ into n overlapped $\mathcal{SCP}_i^k$, $i\epsilon[n]$.
  for $i = 1 : n$
    Form $\mathbf{O}_i^k$ for $\mathcal{SCP}_i^k$ via (10).
    learn $\mathbf{D}_{\mathbf{O}_i^k}$ and $a^1_{\mathbf{O}_i^k}$ for $\mathbf{O}_i^k$ by OSDL.
  end
  Update $\mathbf{\Delta}_k^{(t+1)}$, by concatenating all $\mathbf{D}_{\mathbf{O}_i^k}$ s.
  Update $\mathbf{\Lambda}_k^{(t+1)}$, by concatenating all $a^1_{\mathbf{O}_i^k}$ s.
  Update $\mathbf{SC}_k^{(t+\frac{1}{3})}$ via (23)
  $\mathcal{SC}_k^{(t+\frac{1}{3})} = \texttt{Mat2Ten}^{(3)}\left(\mathbf{SC}_k^{(t+1/3)}\right)$
  for j=1: β_k
  $\mathcal{SC}_k^{(t+\frac{2}{3})}(:,:,j) = \texttt{SpaS}\left(\mathcal{SC}_k^{(t+1/3)}(:,:,j), \text{WT}, \text{N1}\right)$
  $\mathcal{SC}_k^{(t+1)}(:,:,j) = \texttt{Wiener}\left(\mathcal{SC}_k^{(t+2/3)}(:,:,j), \text{NS}\right)$
  end
  $\mathbf{SC}_k^{(t+1)} = \texttt{Ten2Mat}^{(3)}\left(\mathcal{SC}_k^{(t+1)}\right)$
  $\mathbf{\Phi}^{(t+1)}\left(\mathbf{\Delta}_k{}^{(t+1)}, \mathbf{\Lambda}_k{}^{(t+1)}\right) = \mathbf{SC}_k^{(t+1)}$
  Update $\mathbf{V}^{(t+1)}$ via (24)
  end
output $\mathcal{SC}_k^{(f)}$ if $f = iter_{max}$

in 10-nm windows. Some test images of this dataset are shown in Fig. 2.

Dataset B is the 610 × 340-pixeled Pavia University [46], taken by the reflective optic system imaging spectrometer (ROSIS) sensor, consisting of 103 spectral bands that remained after ignoring 12 water vapor absorption bands. We selected the up-right portion of size 256 × 256 pixels from dataset B.

Pixels' intensity values in all the datasets are normalized to the interval [0 1]. To generate the LR versions of the HSIs, we first blur the HR-HSIs by a 3 × 3 averaging filter, then down-sample by a factor of 2 along two spatial dimensions, and finally, add an AWGN signal to the degraded HSI (SNR = 30 dB).

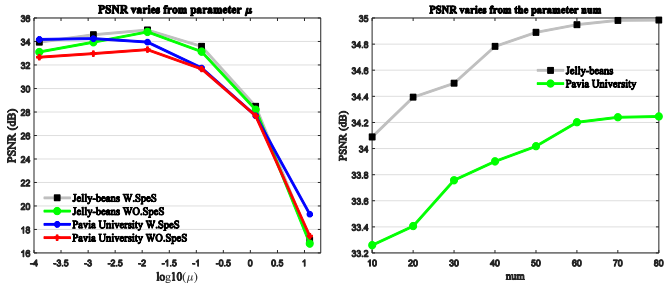
To measure the quantitative accuracy of the super-resolved images, we used four spatial and spectral metrics in this study: 1) mean peak signal-to-noise ratio (MPSNR) is a well-known spatial metric, and the larger the metric's value, the more desirable it is; 2) structural similarity index measure (SSIM) [47] is the other spatial index that determines the super-resolved image similarity to the reference image; like MPSNR, its considerable value is desirable; 3) mean spectral angle mapper (MSAM) [48] measures the spectral distortions of the super-resolved HSI. The smaller the SAM, the more the spectral consistency; and 4) relative dimensionless global error in synthesis (ERGAS) [49]. A lower ERGAS value means a smaller distortion and consequently better reconstruction quality.

B. Initial Setup and Parameters Discussions

In dataset A, we cut the LR-DC to produce three LR-SCs, such that SC1–SC3 consists of 10, 11, and 12 bands, respectively. In dataset B, cutting is done such that the number of LR-SCs is 4 and SC1–SC4 has 30, 31, 31, and 14 spectral channels, respectively. Note that two neighboring SCs overlap in one spectral channel. In each iteration of the SC-SR algorithm, the primary HR-SC is divided into SCPs having a spatial size of 8 × 8 pixels in each band. Furthermore, two adjacent SCPs in each spectral band overlap in four pixels. The spatial size of the 3-D window for clustering co-structural SCPs is 50 × 50, and the number of SCPs gathered in the observations array, i.e., C is set to 30. The n_{low} is selected as equal to $n_{M_{P_i}}$, which means all the frequency components participate in recognizing the co-structural SCPs (see Appendix A). The values of the parameters in Algorithms 1 and 2 are determined experimentally and may vary depending on the datasets, distortion scenario, and the amount of noise. In Algorithm 1, in all the datasets, the balancing factor λ is set to 7×10^{-5}. The value of "R_m" that denotes maximum repetitions in OSDL is set to 5. In Algorithm 2, θ in all the datasets is set to 125×10^{-6}. The penalty parameter (μ) in the objective function, the number of atoms of the dictionary (num) in the OSDL, wavelet type (WT) in the SpaS module, and NS in the AF module are vital parameters discussed below.

The penalty parameter (μ) in the augmented Lagrangian function (19) is the critical factor that directly influences the quality of the super-resolved image. The PSNR curves of the proposed method on *Jelly-Beans* (image selected from dataset A) and dataset B versus $\log(\mu)$ in two modes of with and without exerting SpeS, denoted by **W SpeS**, and **W.O SpeS**, respectively, are plotted in Fig. 3(a). From the plots for the image selected from dataset A, we observe that whether in the **W.O SpeS** mode or **W SpeS** mode, the optimal μ for PSNR is 125×10^{-4}. Thus, we set $\mu = 125 \times 10^{-4}$ for *Jelly-Beans* image.

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(a) (b)

Fig. 3. PSNR varies from parameters: (a) log10 (μ) and (b) num.

TABLE I

VARIATION IN PSNR WITH RESPECT TO SOME WTS IN SPAS MODULE AND VARIOUS DIMENSIONS OF NS IN AF MODULE

WT in `Spas`	Haar	Sym4	Sym18	Sym26	Sym32
Jelly-Beans in Dataset A	30.22	33.97	34.86	34.98	35.01
Time/It (s)	0.80	0.87	1.10	3.48	22.81
Dataset B	31.01	33.76	34.21	34.24	34.28
Time/It (s)	0.74	0.84	1.25	3.64	23.63
NS in `AF`	[3 3]	[5 5]	[7 7]	[9 9]	-
Jelly-Beans in Dataset A	34.98	34.85	34.28	33.50	-
Dataset B	34.24	33.62	32.26	31.39	-

Observing the curves corresponding to dataset B, we can see that in the **W.O SpeS** mode, the optimal values of PSNR occur at $\mu = 125 \times 10^{-4}$, while in **W SpeS** mode, $\mu = 125 \times 10^{-5}$ provides the best values of this metric. According to these results, we set $\mu = 125 \times 10^{-5}$ for dataset B.

The number of dictionary atoms (num) in OSDL is the other crucial factor in the final output of the SR algorithm. The plots of the PSNR varying from the num for *Jelly-Beans* and dataset B are depicted in Fig. 3(b). As can be observed from Fig. 3(b), for both the datasets, the PSNR index is improved as the num varies from 10 to 60, and then it relatively stays constant as the num increases further. However, we see that PSNR has its best value at num = 80; hence, we set it to 80.

WT in SpaS and NS in AF are the parameters that play vital roles in the spatio-spectral resolution of the image resulting from our model. Table I illustrates the variation in the PSNR index for some wavelets and various dimensions of the NS, for the *Jelly-Beans* image and dataset B. Moreover, this table shows the time taken per iteration in the SpaS module for each WT. Note that the time values per iteration have been calculated for SCs with sizes $512 \times 512 \times 12$ and $256 \times 256 \times 31$ in datasets A and B, respectively. Table I shows that when WT varies from sym26 to sym32, the PSNR for the two datasets enhances only slightly. On the other hand, `SpaS` takes much longer when WT is set to Sym32; therefore, Sym26 seems the more logical choice for WT.

In the SpaS module, in addition to WT, the level of the wavelet transform ($N1$) must be determined. For all the datasets and degradation scenarios, the first-level decomposition is applied, i.e., $N1$ is set to 1.

From Table I, we can observe that for the image chosen from dataset A (*Jelly-Beans*) and dataset B, NS = [3 3] provides the optimal values of index PSNR. Thus, we set NS to [3 3] when *Jelly-Beans* and dataset B are the proposed

TABLE II

NS FOR WIENER FILTER, WT IN THE SPES, AND μ FOR SOME TEST IMAGES FOR DATASET A [45] SHOWN IN FIG. 2

Parameter / Image	μ	**NS** in `AF`	**WT** in `SpeS`
(a)	525×10^{-4}	[7 7]	-
(b)	125×10^{-4}	[5 5]	Sym8
(c)	125×10^{-4}	[3 3]	Sym8
(d)	125×10^{-4}	[3 3]	Sym8
(e)	125×10^{-4}	[5 5]	Sym8
(f)	125×10^{-4}	[5 5]	-
(g)	525×10^{-4}	[5 5]	-
(h)	125×10^{-4}	[5 5]	-

TABLE III

AVERAGE QUANTITATIVE METRICS OF RIVAL METHODS ON DATASET A [45]. DEGRADATION SCENARIO: 3×3 AVERAGING FILTER, DECIMATION FACTOR OF 2, AND AWGN 30 dB

Methods / Metrics	MPSNR	MSSIM	MSAM	ERGAS
Bicubic	34.4243	0.9476	9.2589	10.9471
SSSM [21]	35.6624	0.9561	8.8229	9.2109
A+NR [27]	36.8005	0.9602	7.9112	8.6527
MAP [22]	37.1933	0.9638	7.1551	8.2160
Proposed	**39.5045**	**0.9705**	**5.8921**	**6.5521**

SR algorithm's input images. The optimal values of the penalty parameter (μ) and NS in the AF module for the test images shown in Fig. 2 are given in the second and third columns of Table II, respectively.

In dataset A, as stated in Section II-E, we use 1-D wavelet-based smoothing pixel by pixel for the SpeS module, where the first-level decomposition is used, i.e., $N2 = 1$. Furthermore, according to the fourth column of Table II, *Sym8* is the wavelet used in SpeS for those images to which this module is applied.

In dataset B, for the SpeS module, the command cmddenoise(sig,wname,level) is used pixel by pixel, in which sig, wname, and level are set to the spectral vector of each pixel, sym8, and 3, respectively.

C. Compared Methods

The proposed model is compared with the bicubic interpolation and the state-of-the-art methods, SSSM [21], A + NR [27], MAP [22], and MW-3D-CNN [28]. It should be noted that the deep-learning-based method [28] is excluded from the experiment on dataset A. The reason is that the techniques based on deep learning require many imagery samples to train neural networks; on the other hand, in this article, all the 32 images contained in dataset A with their full-size dimensions, i.e., $512 \times 512 \times 31$ pixels, are used as the test data. Thus, no imagery samples would remain from this dataset to be used as the training samples. However, in dataset B, we take a portion of size 256×256 pixels as the test data and use other parts that do not overlap the test area as the training samples. Hence, all the competitors are included in the experiment on dataset B.

D. Experimental Results on Synthetic LR-Data

The average values of the quantitative comparisons between the proposed method and the others for datasets A and B are given in Tables III and IV, respectively. The two tables

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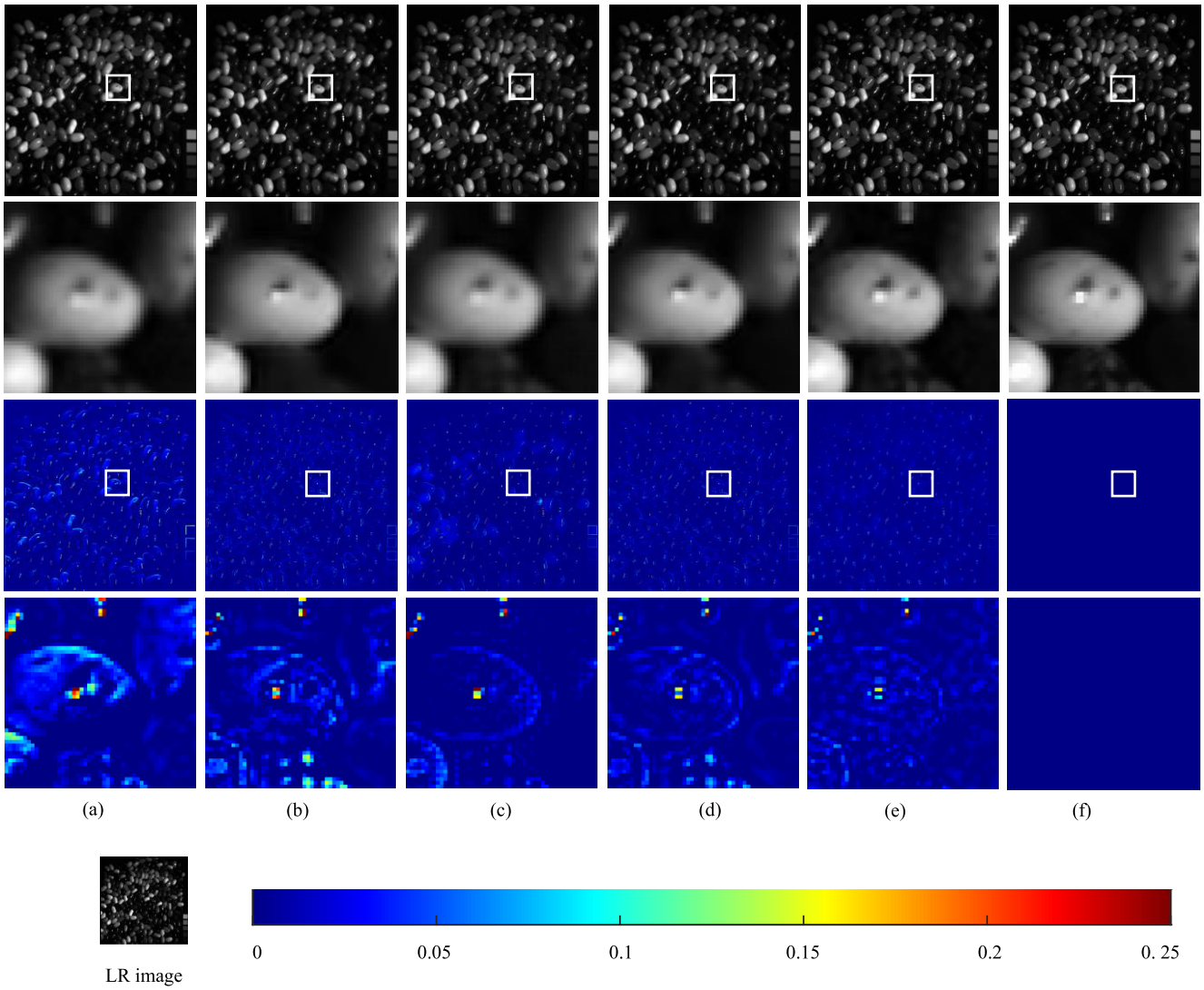


Fig. 4. (First row) Super-resolved images for the "*Jelly-Beans*" from dataset A at band 15. (Second row) Zoomed-in versions of the part specified by white squares in the images of the first row. (Third row) Error color maps (differences between the ground truth and the super-resolved images) for the whole scene. (Fourth row) Error color maps for the selected area of the scene. (a) Bicubic. (b) SSSM [21]. (c) A + NR [27]. (d) MAP [22]. (e) Proposed. (f) Ground truth. The low-resolution image is given beside the color bar.

TABLE IV

AVERAGE QUANTITATIVE METRICS OF RIVAL METHODS ON DATASET B [46]. DEGRADATION SCENARIO: 3 × 3 AVERAGING FILTER, DECIMATION FACTOR OF 2, AND AWGN 30 dB

Methods \ Metrics	MPSNR	MSSIM	MSAM	ERGAS
Bicubic	29.3628	0.8371	4.6565	10.2568
SSSM [21]	30.4013	0.8503	4.4892	8.3027
A+NR [27]	31.7821	0.8610	4.3101	7.5139
MAP [22]	32.1794	0.8976	4.2719	7.3831
MW-3D CNN [28]	33.7535	0.9118	4.0256	6.5168
Proposed	**34.2461**	**0.9293**	**3.5542**	**5.9322**

show that the proposed method proves superior to the other compared methods in all the objective metrics.

For visual comparison, the super-resolution results of the competing methods and the ground-truth version at band 15 for *Jelly-Beans* (the image selected from dataset A) are provided in Fig. 4. The first row of this figure shows the ground-truth image and the super-resolved HSIs attained by the various methods. Moreover, in the images of the first row, we see white squares marking a particular region of the scene. The zoomed-in versions of this selected area are demonstrated in the second row. Furthermore, the error color maps for the whole image and the magnified version of the selected region are shown in the third and fourth rows, respectively. The error color map visually denotes the difference between the super-resolved and ground-truth versions. Likewise, Fig. 5 illustrates the visual results of the competing methods at band 65 for dataset B. From these two figures, it can be perceived that the images super-resolved by the proposed method have fewer errors and artifacts than the others; furthermore, our method outperforms the others in preserving edges and high-frequency information.

We have also conducted another experiment to compare the proposed method with the state-of-the-art single-frame deep learning method G-RDN [29] and a well-known fusion-based approach CSTF [13]. In this comparison, to generate

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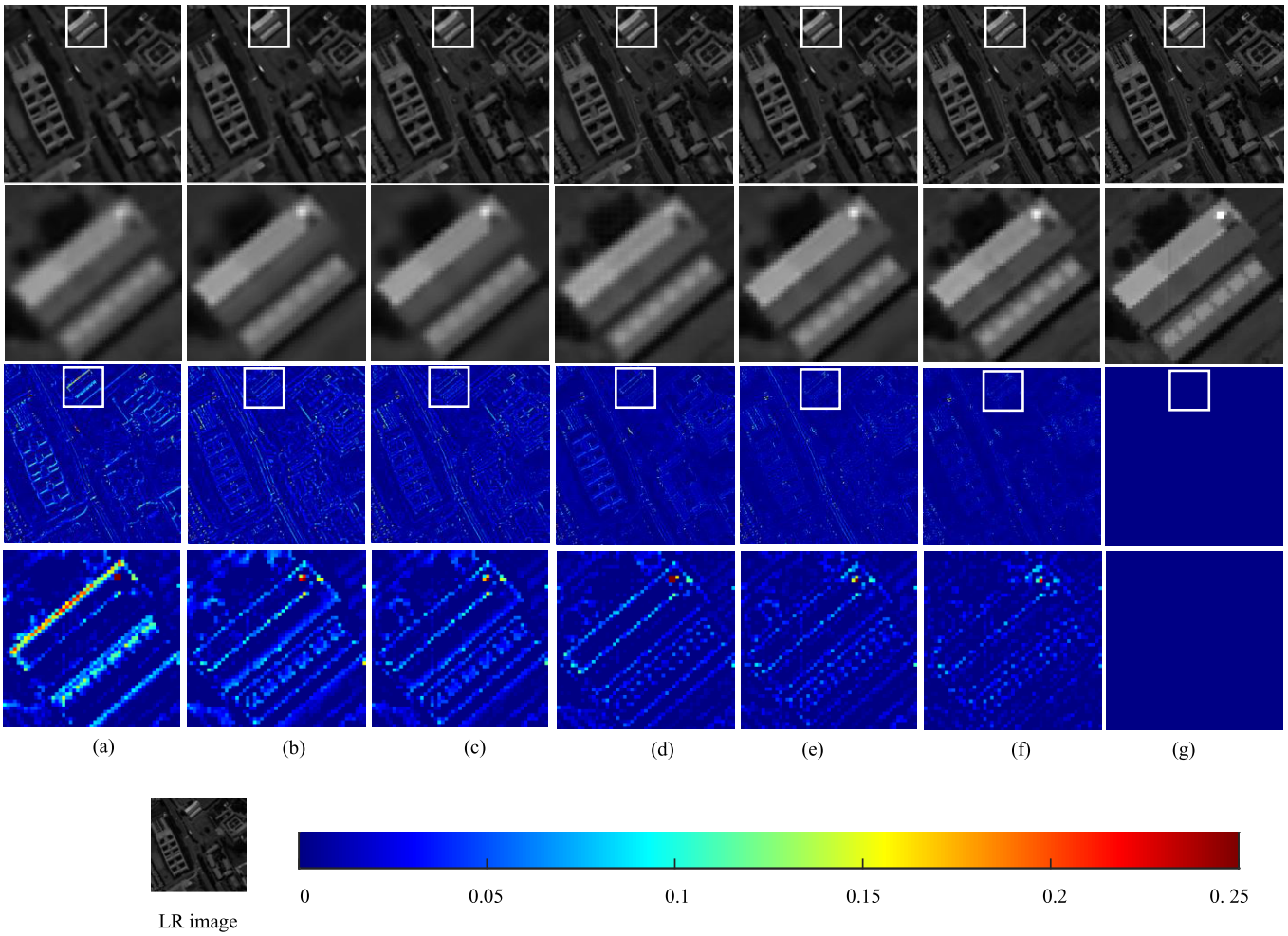


Fig. 5. (First row) Super-resolved images for dataset B at band 65. (Second row) Zoomed-in versions of the part specified by white squares in the images of the first row. (Third row) Error color maps (differences between the ground truth and the super-resolved images) for the whole scene. (Fourth row) Error color maps for the selected area of the scene. (a) Bicubic. (b) SSSM [21]. (c) A + NR [27]. (d) MAP [22]. (e) MW-3D-CNN [28]. (f) Proposed. (g) Ground truth. The low-resolution image is given beside the color bar.

LR-HSIs, we have exerted a degradation scenario commonly used to assess fusion-based strategies. First, an 8×8 Gaussian blurring filter with a standard deviation of 2 is applied to HR-HSIs, and then blurred HSIs are down-sampled with the decimation factor of 8 along both the spatial dimensions (vertical and horizontal). The HSIs *stuffed-toy*, *Balloons*, and *fake and real apples* have been selected from dataset A as the test images. In this scenario, the parameters of our method are set as follows:

$(\mu, \theta) = (125 \times 10^{-6},\ 125 \times 10^{-8})$, WT in SpaS is set to *Sym26*, NS = [13 13], and finally, the SpeS module is not applied. In G-RDN [29], the remaining 29 in dataset A are intended for the training and validation sets, other than the three images used as the test set. Table V reports the average indices of test approaches on these three HSIs selected from dataset A. It is clear that CSTF [13] by far outperforms the proposed method and G-RDN. The reason is that CSTF is a fusion-based method that uses the high spatial resolution multispectral versions of the original scenes. Contrary to CSTF, our approach and G-RDN only use the single low-resolution frame of the taken HSI. However, G-RDN uses 29 HR-HSIs to set up its model's requisites (training and validation phases), but the proposed method does not use any image other than the captured LR-HSI. Nevertheless, from the results in Table V, we can find that our model performs better than G-RDN.

TABLE V

AVERAGE QUANTITATIVE METRICS OF RIVAL METHODS ON THREE IMAGES OF DATASET A [45]. DEGRADATION SCENARIO: 8×8 GAUSSIAN FILTER S.D = 2 AND DECIMATION FACTOR OF 8

Methods \ Metrics	MPSNR	MSSIM	MSAM	ERGAS
Bicubic	27.0174	0.9125	4.5035	8.7314
G-RDN [29]	30.1271	0.9341	4.1212	6.0309
Proposed	31.4719	0.9439	3.2194	3.3120
CSTF [13]	**41.5326**	**0.9808**	**2.6129**	**1.0289**

E. Ablation Study on Features and Modules

In this section, we present ablation studies conducted on the features (**F1–F5**) to validate their influence on the results of the proposed SR algorithm.

The image *Jelly-Beans* from datasets A and B are analyzed under the first degradation scenario we used. Tables VI and VII

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TABLE VI

ABLATION STUDY ON THE FEATURES (**F2–F5**) AND THEIR CORRESPONDING MODULES FOR EACH SC OF JELLY-BEANS FROM DATASET A IN DISTORTION SCENARIO: DS = **2**, **3** × **3** AVERAGING FILTER, AND AWGN **30** dB

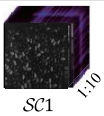
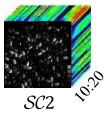
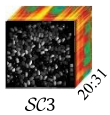
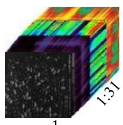
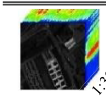
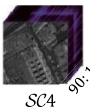
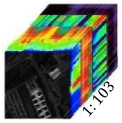
F2, F3 (OSDL)		OFF	ON	ON	ON
F5 (Spas)		ON	OFF	ON	ON
F4 (AF)		ON	ON	OFF	ON
 $SC1$ 1:10	MPSNR	34.5292	34.2220	33.2063	**34.5966**
	MSSIM	0.9534	0.9526	0.9178	**0.9582**
	MSAM	8.3018	8.4157	11.5984	**7.5339**
	ERGAS	9.6891	9.8389	11.0152	**9.5291**
	Iter_{Max}	10	70	2	10
Time/Iteration (s)		**3.68**	166.73	170.03	170.22
 $SC2$ 10:20	MPSNR	35.1196	34.8546	33.9954	**35.2286**
	MSSIM	0.9529	0.9521	0.9199	**0.9586**
	MSAM	15.4506	15.5943	20.2938	**11.7848**
	ERGAS	7.6879	7.9561	8.7787	**7.5550**
	Iter_{Max}	10	70	2	10
Time/Iteration (s)		**4.39**	179.79	183.73	183.96
 $SC3$ 20:31	MPSNR	34.5035	34.3325	33.4370	**34.5861**
	MSSIM	0.9558	0.9559	0.9282	**0.9593**
	MSAM	8.7380	8.8613	11.6395	**7.2711**
	ERGAS	5.0595	5.1137	5.6667	**4.9579**
	Iter_{Max}	10	70	2	10
Time/Iteration (s)		**4.59**	190.71	194.82	195.06
 $\mathcal{X}^{(f+\frac{1}{2})}$ 1:31	MPSNR	34.7264	34.4717	33.5118	**34.8200**
	MSSIM	0.9563	0.9543	0.9224	**0.9604**
	MSAM	7.8314	8.1599	10.8370	**6.9492**
	ERGAS	7.4917	7.6800	8.5575	**7.3753**

TABLE VII

ABLATION STUDY ON THE FEATURES (**F2–F5**) AND THEIR CORRESPONDING MODULES FOR EACH SC OF DATASET B IN DISTORTION SCENARIO: DS = **2**, **3** × **3** AVERAGING FILTER, AND AWGN **30** dB

F2, F3 (OSDL)		OFF	ON	ON	ON
F5 (SpaS)		ON	OFF	ON	ON
F4 (AF)		ON	ON	OFF	ON
 $SC1$ 1:30	MPSNR	33.5110	32.9501	32.8590	**33.6284**
	MSSIM	0.8692	0.8584	0.8439	**0.8721**
	MSAM	9.5598	10.2710	10.8021	**9.1967**
	ERGAS	8.5573	9.4526	9.2218	**8.4557**
	Iter_{Max}	10	30	10	10
Time/Iteration (s)		**4.33**	110.67	114.8	114.9
 $SC2$ 30:60	MPSNR	33.3289	32.7036	32.7085	**33.4107**
	MSSIM	0.8985	0.8878	0.8749	**0.9000**
	MSAM	7.3402	8.1501	8.6584	**6.7839**
	ERGAS	7.1810	8.1553	7.7402	**7.1241**
	Iter_{Max}	10	30	10	10
Time/Iteration (s)		**4.52**	115.52	119.82	119.93
 $SC3$ 60:90	MPSNR	32.5560	31.7919	32.0809	**32.5834**
	MSSIM	0.9027	0.8947	0.8878	**0.9050**
	MSAM	3.9468	4.5119	4.4504	**3.8616**
	ERGAS	5.5192	6.2363	5.8730	**5.5002**
	Iter_{Max}	10	30	10	10
Time/Iteration (s)		**4.52**	115.52	119.82	119.93
 $SC4$ 90:103	MPSNR	31.7395	30.9415	31.3942	**31.7490**
	MSSIM	0.8998	0.8893	0.8851	**0.9010**
	MSAM	2.8521	3.3419	3.2806	**2.8394**
	ERGAS	4.4368	4.8943	4.6183	**4.4330**
	Iter_{Max}	10	30	10	10
Time/Iteration (s)		**2.69**	53.36	55.97	56.01
 $\mathcal{X}^{(f+\frac{1}{2})}$ 1:103	MPSNR	32.9036	32.3948	32.4968	**33.0367**
	MSSIM	0.8895	0.8826	0.8707	**0.8940**
	MSAM	5.2109	5.7036	5.7835	**5.0372**
	ERGAS	6.8966	7.4535	7.3983	**6.8181**

demonstrate the impact of **F2–F5** on the quantitative metrics, running time, and Itermax for datasets A and B, respectively; these tables report indices' values for SCs and the HR-DC resulting from merging them.

Table VIII shows the effect of **F1** on the final super-resolved HR-DC for both datasets A and B. The results of ablation experiments are as follows. Note that we compare the deactivated mode of each module with the case all the modules are activated (the last column in the tables).

As can be seen, in the *Jelly-Beans* image, learning dictionary atoms and sparse coefficients, the most time-consuming module of the model, positively impacts all the assessment metrics, especially SAM, which is improved approximately 0.9° in the HR-DC. As a result, when the OSDL module is disabled, the value of SAM increases, i.e., the spectral consistencies of SCs and HR-DC decrease. Therefore, as OSDL has led to a noticeable improvement in the spectral structure and simultaneously keeps the other indices high, we can interpret that this module has well played the role of injecting spatio-spectral information. As can be observed, disabling OSDL in dataset B negatively affects all the indices in the SR algorithm; however, each iteration's running takes longer when this module is activated. SpaS deactivation causes all metrics' deterioration and increases the number of iterations; consequently, the algorithm's total running time grows longer for datasets A and B. In both the images, in the AF-deactivated mode, the quantitative metrics, especially SAM, have been diminished; moreover, ignoring AF, the best indices' values occur in the second iteration, i.e., Itermax = 2. From Table VIII, we can see that in the *Jelly-Beans* image, applying SpeS to HR-DC gives rise to PSNR, SSIM, SAM, and ERGAS improvement, approximately 0.18, 0.09, 1.34°, and 0.17, respectively. For some images in dataset A, because of the tradeoff between spatial and spectral resolutions, SpeS does not simultaneously influence all the metrics positively. In dataset B, SpeS leads to ameliorating all the quality metrics remarkably. In addition, SAM and ERGAS have been improved by 1.5° and 0.9, respectively.

TABLE VIII

ABLATION STUDY ON THE FEATURE (**F1**) AND THE CORRESPONDING MODULE FOR HR-DC OF JELLY-BEANS IMAGE AND DATASET B

Dataset		*Jelly-Beans* in Dataset A		Dataset B	
F1 (SpeS)		OFF	ON	OFF	ON
$\mathcal{X}^{(f)}$	MPSNR	34.8200	**34.9844**	33.0367	**34.2461**
	MSSIM	0.9604	**0.9692**	0.8940	**0.9293**
	MSAM	6.9492	**5.6137**	5.0372	**3.5542**
	ERGAS	7.3753	**7.2042**	6.8181	**5.9322**
Time (s)		0	294.69	0	320.10

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TABLE IX
AVERAGE COMPUTATIONAL TIME (IN SECONDS) OF RIVAL METHODS ON DATASET A (OF SIZE 512 × 512 × 31) AND DATASET B (OF SIZE 256 × 256 × 103)

Method / Dataset	Bicubic	SSSM [21]	A+NR [27]	MAP [22]	MW-3D-CNN [28]	Proposed
Dataset A	0.08	41	11.3	3036	4.6	2524
Dataset B	0.06	25	8.7	518	2.2	1710

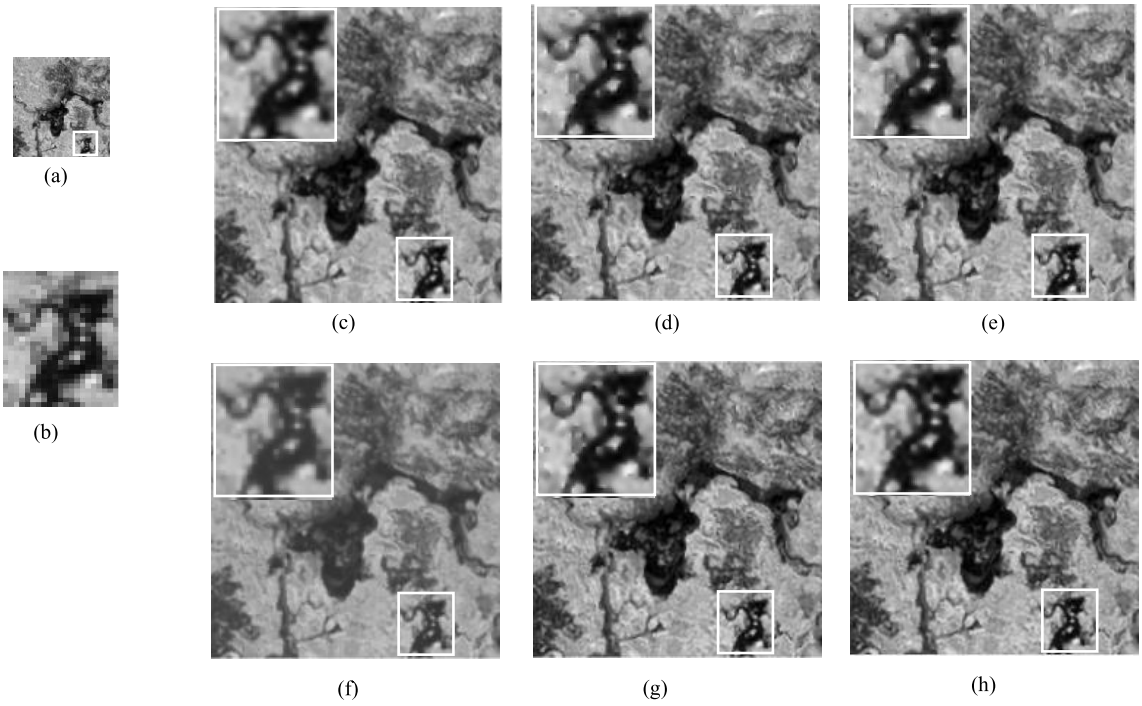


Fig. 6. HR-HSIs (of Hyperion data at band 35) super-resolved by competing methods. (a) LR-HSI. (b) Zoomed-in version of the selected area in LR-HSI. (c) Bicubic. (d) SSSM [21]. (e) A + NR [27]. (f) MAP [22]. (g) MW-3D-CNN [28]. (h) Proposed.

F. Time-Efficiency and Computational Complexity

Table IX shows the time efficiency of the compared models for the two datasets under the first degradation scenario. MAP [22] takes more time in dataset A than the other methods, whereas in dataset B, our method is the most time-consuming one. As mentioned in the section of the ablation study, the OSDL module, whose task is injecting information from co-structural SCPs into each other, consumes the most time in the proposed model. Ignoring the bicubic method, which is the most time-efficient compared with competing methods, we can see that A + NR [27] and MW-3D-CNN [28] are the fastest techniques. However, it is noted that these two approaches have training phases, which take several hours [28].

In Algorithm 1, the complexity of the innermost loop is $O\left(4\beta_k m_p n_p C + \beta_k m_p n_p\right)$. Thus, for the "num" atoms "R_m" and computational complexity in learning dictionary and sparse coefficients for the observations, the array of each SCP would be $O\left(R_m \times \text{num} \times (4C+1)\beta_k m_p n_p\right)$.

In Algorithm 2, based on [10], the computational complexity of updating $\mathbf{U}_k$, whose final solution is given in Appendix B in (33), is $O\left(\sqrt{\gamma}(\beta_k m_X n_X)\log m_X n_X\right)$ where γ is the condition number of $\mathbf{Q}$ in (35). Since $\gamma \epsilon O(m_X n_X)$, the computational complexity for updating $\mathbf{U}_k$ would be $O\left(\beta_k (m_X n_X)^{3/2} \log m_X n_X\right)$. Finally, we used a laptop with CPU Core i7, 2.4 GHz, and 8-GB DDR3 L memory to test the proposed method.

G. Real HSI Experiment

This section presents an experimental study to assess the proposed model when its input is a real LR-HSI. The NASA Earth Observing-1 (EO-1) satellite captured a data series over Okavango Delta, Botswana (2001–2004). This real HSI was collected by the Hyperion sensor installed on EO-1 [46]. The HSI taken by this sensor is of 30-m spatial resolution and contains 240 spectral bands ranging from 400 to 2500 nm with 10-nm widths. After removing the water absorption bands, which were uncalibrated and noisy, the resulting HSI, with a spatial size of 1476 × 256 in 90 spectral channels, is considered the real data from which we select a region of size 128 × 128 × 90 to conduct our experiment. In a single-image-based approach, it is assumed that there is no auxiliary high-resolution image of the scene like MSI, based on which we can determine the decimation factor and estimate the blurring kernel using techniques such as in [5]. Therefore, the degradation operator practically is unknown. However, upscaling real low-resolution data is commonly used to test the effectiveness of a super-resolution model; thus, using the proposed and competing methods, we upscale the available

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single LR-HSI by a factor of 3. Accordingly, we suppose that the LR-HSI of size 128 × 128 × 90 has been generated by averaging over the 3 × 3 disjoint spatial blocks of a hypothetical high-resolution HSI of size 384 × 384 × 90. For upscaling the real data, the proposed method's parameters are tuned as follows: $(\mu, \theta) = (125 \times 10^{-6},\ 125 \times 10^{-8})$, Sym26 is selected for the WT in the SpaS module, and NS in the AF module is set to [3 3]. Because of the lack of a ground-truth version, the same protocol used in [28] was used to train its neural network. From Fig. 6, which shows the super-resolved HR-HSIs at band 35 for Hyperion data, we see that all the methods, especially the proposed one, enhance the spatial resolution of input LR-HSI well. However, although the Bayesian-based algorithm used in [22] can improve the spatial resolution well, the super-resolved image resulting from this technique is over-smooth due to the fact that it used regularizers focused on reconstructing regions with high homogeneity. As can be seen, the image upscaled by the proposed method contains more information (both low- and high-frequency information) compared with the input scene and images reconstructed by other competitors.

IV. Conclusion

In this article, we present a novel HSI-SR technique, which obtains a spatio-spectrally HR-HSI from only a single LR-HSI. The proposed approach is based on fully using HSIs' characteristics without using complex regularizers in the final cost function. The LR-HSI-SR problem is transformed into several identical LR-SC-SR problems solved by a unique SC-SR algorithm. This algorithm is implemented as a modular model with two main parts. The first exploits prior information of the high spectral resolution HSI, while the second uses wavelet-based SpaS and AF. Finally, after merging all the super-resolved SCs and providing an HR-HSI, the SRFs are denoised via wavelet-based smoothing in the spectral dimension. The simulation results on two synthetic LR datasets and one real LR-HSI show that the proposed method outperforms the other competing state-of-the-art methods and has fewer artifacts.

Appendix A

As shown in Fig. 7, in a typical image, we can observe patches being highly similar to the rotated versions of each other. In such cases, the criteria Euclidean norm used in [50] would not be able to measure the nonlocal similarity well. For example, if the white and gray parts of three magnified patches, $p1$, $p2$, and $p3$, are mapped to 0 and 0.5, respectively, the patches can be shown by three matrices, as given in Fig. 7. In this case, the minimum Euclidean distance reports $p3$ as the patch similar to $p1$ since

$$\|p1 - p3\|_F^2 < \|p1 - p2\|_F^2 \tag{26}$$

while it is clear that $p1$ is more similar to $p2$ than to $p3$.

Inspired by Zuo et al. [51], we use a rotation-invariant criterion to cluster the co-structural SCPs as follows:

$$\begin{aligned} &p_i = \mathrm{SCP}_i(:), \quad \mathrm{M}_{p_i} = |\mathrm{fft}(p_i)|, \ n_{\mathrm{M}_{p_i}} = \mathrm{length}(\mathrm{M}_{p_i}) \\ &\mathrm{M}_{p_i}^{\mathrm{low}} = \mathrm{M}_{p_i}(1 : n_{\mathrm{low}}) \end{aligned} \tag{27}$$

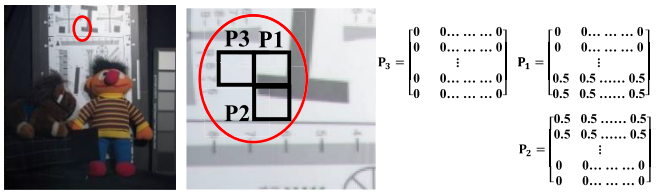


Fig. 7. Three similar patches and their matrix representation, if the white and gray parts are mapped onto 0 and 0.5, respectively.

where fft(·) is the built-in function in MATLAB to compute a vector's FFT. In addition, |.| and length(·) return the magnitude and size of a vector, respectively, and n_{low} denotes the number of low-frequency components. Hence, the criterion to recognize co-structural SCPs is as follows.

If

$$m_1 = \left\|\mathrm{M}_{p_1}^{\mathrm{low}} - \mathrm{M}_{p_2}^{\mathrm{low}}\right\|_2, \quad m_2 = \left\|\mathrm{M}_{p_1}^{\mathrm{low}} - \mathrm{M}_{p_3}^{\mathrm{low}}\right\|_2 \tag{28}$$

then

$$\{\text{matched SCP to } \mathrm{SCP}_1\} = \begin{cases} \mathrm{SCP}_2, & \text{if } m_1 < m_2 \\ \mathrm{SCP}_3, & \text{if } m_1 > m_2. \end{cases} \tag{29}$$

There are two advantages to using $\mathrm{M}_{p_i}^{\mathrm{low}}$; first, the patch rotation does not impact it, and second, as stated, the low-frequency components contain the major information of the image, and the signal-to-noise ratio (SNR) in these components is high. Therefore, the comparison among $\mathrm{M}_{p_i}^{\mathrm{low}}$s to cluster co-structural SCPs is a rotation-invariant and noise-resistant criterion.

Appendix B

In this appendix, we first give the mathematical definitions of the entrywise matrix l_1 norm and the spectral total variation operator used in (16). Then we present the details of solving the optimization problems (20) and (21).

Definition 1: The entrywise matrix l_1 norm, denoted by $\|\cdot\|_{1,1}$, operates on a matrix as its argument and returns the sum of the absolute values of all the entries of that matrix, e.g., if we have $\mathbf{\Psi} \in \mathbb{R}^{m \times n}$, then

$$\|\mathbf{\Psi}\|_{1,1} = \|\mathbf{\Psi}(:)\|_1 = \sum_{i=1}^{m}\sum_{j=1}^{n}|\psi_{i,j}|. \tag{30}$$

Definition 2: The spectral total variation operator, denoted by $\mathbf{T} \in \mathbb{R}^{\beta_k \times \beta_k}$ in (16), is defined as below:

$$\mathbf{T} = \begin{cases} T_{i,i} = -1, & \forall\ 1 \le i \le \beta_k \\ T_{i,i+1} = 1, & \forall\ 1 \le i \le \beta_k - 1 \\ 0, & \text{O.W.} \end{cases} \tag{31}$$

The Final Solutions to Optimization Problems (20) and (21).

We rewrite the two optimization problems here, and then present their final solutions.

The updating subproblem (20) can be rewritten as follows:

$$\begin{aligned} \mathbf{U}_k^{(t+1)} = \mathrm{argmin}_{\mathbf{U}_k} &\frac{1}{2}\|\mathbf{U}_k\mathbf{Z} - \mathbf{Y}_k\|_F^2 + \theta\left\|\mathbf{T}\mathbf{\Phi}^{(t)}\left(\mathbf{\Delta}_k{}^{(t)}, \mathbf{\Lambda}_k{}^{(t)}\right)\right\|_{1,1} \\ &+ \mu\left\|\mathbf{U}_k - \mathbf{\Phi}^{(t)}\left(\mathbf{\Delta}_k{}^{(t)}, \mathbf{\Lambda}_k{}^{(t)}\right) + \frac{\mathbf{V}^{(t)}}{2\mu}\right\|_F^2. \end{aligned} \tag{32}$$

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The solution of $\mathbf{U}_k^{(t+1)}$ is

$$\mathbf{U}_k^{(t+1)} = \mathbf{qQ} \tag{33}$$

where

$$\mathbf{q} = \left(\mathbf{Y}_k\mathbf{Z}^* + 2\mu\left(\mathbf{\Phi^{(t)}}\left(\mathbf{\Delta}_k{}^{(t)}, \mathbf{\Lambda}_k{}^{(t)}\right) - \frac{\mathbf{V}^{(t)}}{2\mu}\right)\right), \tag{34}$$

$$\mathbf{Q} = (\mathbf{ZZ}^* + 2\mu\mathbf{I}_{\mathbf{d}(m_x n_x \times m_x n_x)})^{-1} \tag{35}$$

where $\mathbf{I}_{\mathbf{d}(m_x n_x \times m_x n_x)}$ is the identity matrix of size $m_x n_x \times m_x n_x$.

The updating subproblem (21) can be rewritten as follows:

$$\begin{aligned}\mathbf{\Phi}^{(t+\frac{1}{2})}\left(\mathbf{\Delta}_k{}^{(t)}, \mathbf{\Lambda}_k{}^{(t)}\right) = \operatorname{argmin}_{\Phi(\mathbf{\Delta}_k,\mathbf{\Lambda}_k)} &\frac{1}{2}\left\|\mathbf{U}_k{}^{(t+1)}\mathbf{Z} - \mathbf{Y}_k\right\|_F^2 \\ &+ \theta\|\mathbf{T}\Phi(\mathbf{\Delta}_k, \mathbf{\Lambda}_k)\|_{1,1} \\ &+ \mu\left\|\mathbf{U}_k{}^{(t+1)} - \mathbf{\Phi}(\mathbf{\Delta}_k, \mathbf{\Lambda}_k) + \frac{\mathbf{V}^{(t)}}{2\mu}\right\|_F^2.\end{aligned} \tag{36}$$

The solution of $\mathbf{\Phi}^{(t+(1/2))}\left(\mathbf{\Delta}_k{}^{(t)}, \mathbf{\Lambda}_k{}^{(t)}\right)$ is

$$\mathbf{\Phi}^{(t+\frac{1}{2})}\left(\mathbf{\Delta}_k{}^{(t)}, \mathbf{\Lambda}_k{}^{(t)}\right) = \mathbf{T}^* \operatorname{softhr}\left[\left(\mathbf{T}^{-1}\right)^*\left(\mathbf{U}_k^{(t+1)} + \frac{\mathbf{V}^{(t)}}{2\mu}\right), \frac{\theta}{2\mu}\right] \tag{37}$$

where $\mathbf{T}^*$ is the Hermitian transpose of $\mathbf{T}$. Moreover, softhr[a, θ] is the soft-threshold function [39] defined as follows:

$$\operatorname{softhr}[\mathrm{a}, \theta] = \begin{cases} \mathrm{a} - \theta, & \mathrm{a} > \theta \\ 0, & |\mathrm{a}| \le \theta \\ \mathrm{a} + \theta, & \mathrm{a} < -\theta. \end{cases} \tag{38}$$

Appendix C

The pixelwise Wiener filter [41] used in this article is briefly described as follows. As seen in the text, a pixel at coordinates (i,j) in the kth band of an HSI would be denoted by $\mathbf{X}^k(i, j)$. The Wiener filter calculates $\mathbf{X}^k_{\mathrm{Out}}(i, j)$ as the estimation of $\mathbf{X}^k_{\mathrm{Un}}(i, j)$. The subscripts "Out," "Un," and "In" denote the output (estimated), undegraded, and input (degraded) images, respectively. For a pixel at coordinates (i,j), a neighborhood of size NS = $[2m_{\mathrm{nei}} + 1,\ 2n_{\mathrm{nei}} + 1]$ is considered, then by estimating the local means and variances, the pixel (i,j) of the undegraded image is estimated as follows:

$$\hat{\eta}_{\mathbf{X}^k_{\mathrm{Un}}}(i, j) = (\mathrm{PS}) \times \sum_{\substack{k_1 = \\ i-m_{\mathrm{nei}}}}^{i+m_{\mathrm{nei}}} \sum_{k_2 = j-n_{\mathrm{nei}}}^{j+n_{\mathrm{nei}}} \mathbf{X}^k_{\mathrm{In}}(k_1, k_2) \tag{39}$$

$$\hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{In}}}(i, j) = (\mathrm{PS}) \times \sum_{\substack{k_1 = \\ i-m_{\mathrm{nei}}}}^{i+m_{\mathrm{nei}}} \sum_{\substack{k_2 = \\ j-n_{\mathrm{nei}}}}^{j+n_{\mathrm{nei}}} \left(\mathbf{X}^k_{\mathrm{In}}(k_1, k_2) - \hat{\eta}_{\mathbf{X}^k_{\mathrm{Un}}}(k_1, k_2)\right)^2 \tag{40}$$

$$\hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{Un}}}(i, j) = \begin{cases} \hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{In}}}(i, j) - \sigma_v^2, & \text{if } \hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{In}}}(i, j) > \sigma_v^2 \\ 0, & \text{O.W} \end{cases} \tag{41}$$

$$\begin{aligned}\hat{\mathbf{X}}^k_{\mathrm{Un}}(i, j) = \mathbf{X}^k_{\mathrm{Out}}(i, j) &= \mathtt{Wiener}\left(\mathbf{X}^k_{\mathrm{In}}(i, j)\right) \\ &= \hat{\eta}_{\mathbf{X}^k_{\mathrm{Un}}}(i, j) + \frac{\hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{Un}}}(i, j)}{\hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{Un}}}(i, j) + \sigma_v^2} \\ &\quad \times \left(\mathbf{X}^k_{\mathrm{In}}(i, j) - \hat{\eta}_{\mathbf{X}^k_{\mathrm{Un}}}(i, j)\right)\end{aligned} \tag{42}$$

where PS = $(1/(2m_{\mathrm{nei}} + 1)(2n_{\mathrm{nei}} + 1))$, and η and σ denote the mean and variance, respectively. When the noise variance, σ_v^2, is unknown, the mean of all $\hat{\sigma}^2_{\mathbf{X}^k_{\mathrm{In}}}(i, j)$ is considered as the σ_v^2. Therefore, to obtain the output of this filter, we only need the input image (degraded image) and the NS.

References

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] H. Zhang, L. Zhang, and H. Shen, "A super-resolution reconstruction algorithm for hyperspectral images," *Signal Process.*, vol. 92, no. 9, pp. 2082–2096, Sep. 2012.

[3] Y. A. Zhao, J. Yang, and J. C.-W. Chan, "Hyperspectral imagery super-resolution by spatial–spectral joint nonlocal similarity," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2671–2679, Dec. 2013.

[4] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1–15.

[5] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[6] Y. Fu, Y. Zheng, H. Huang, I. Sato, and Y. Sato, "Hyperspectral image super-resolution with a mosaic RGB image," *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5539–5552, Nov. 2018.

[7] R. Dian, S. Li, L. Fang, T. Lu, and J. M. Bioucas-Dias, "Nonlocal sparse tensor factorization for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4469–4480, Oct. 2020.

[8] J. Xue, Y. Zhao, Y. Bu, W. Liao, J. C. Chan, and W. Philips, "Spatial–spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[9] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[10] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[11] C. Lin, F. Ma, C. Chi, and C. Hsieh, "A convex optimization-based coupled nonnegative matrix factorization algorithm for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1652–1667, Mar. 2018.

[12] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "A truncated matrix decomposition for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 29, pp. 8028–8042, 2020.

[13] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[14] R. Dian and S. Li, "Hyperspectral image super-resolution via subspace-based low tensor multi-rank regularization," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5135–5146, Oct. 2019.

[15] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.

[16] P. Zhou, C. Lu, Z. Lin, and C. Zhang, "Tensor factorization for low-rank tensor completion," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1152–1163, Mar. 2018.

[17] Y. Bu et al., "Hyperspectral and multispectral image fusion via graph Laplacian-guided coupled tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 648–662, May 2020.

[18] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:05:30 UTC from IEEE Xplore. Restrictions apply.

[19] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.
[20] L. Su, S. Zhou, and Y. Yuan, "High spatial resolution image restoration from subpixel-shifted hyperspectral images," *J. Appl. Remote Sens.*, vol. 9, no. 1, Feb. 2015, Art. no. 095093.
[21] X. Xu et al., "Hyperspectral image super resolution reconstruction with a joint spectral–spatial sub-pixel mapping model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1–4.
[22] H. Irmak, G. B. Akar, and S. E. Yuksel, "A MAP-based approach for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2942–2951, Jun. 2018.
[23] H. Wang, W. Li, L. Hu, C. Zhang, and Q. He, "Structural smoothness low-rank matrix recovery via outlier estimation for image denoising," *Multimedia Syst.*, vol. 28, no. 1, pp. 241–255, Feb. 2022.
[24] X. Han, B. Shi, and Y. Zheng, "Self-similarity constrained sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5625–5637, Nov. 2018.
[25] J. Zhang, D. Zhao, and W. Gao, "Group-based sparse representation for image restoration," *IEEE Trans. Image Process.*, vol. 23, no. 8, pp. 3336–3351, Aug. 2014.
[26] J. Li, Q. Yuan, H. Shen, X. Meng, and L. Zhang, "Hyperspectral image super-resolution by spectral mixture analysis and spatial–spectral group sparsity," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 9, pp. 1250–1254, Sep. 2016.
[27] R. Timofte, V. De Smet, and L. Van Gool, "A+: Adjusted anchored neighborhood regression for fast super-resolution," in *Proc. Asian Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 111–126.
[28] J. Yang, Y. Q. Zhao, J. C. W. Chan, and L. Xiao, "A multi-scale wavelet 3D-CNN for hyperspectral image super-resolution," *Remote Sens.*, vol. 11, no. 13, p. 1557, Jun. 2019.
[29] M. Zhao, J. Ning, J. Hu, and T. Li, "Hyperspectral image super-resolution under the guidance of deep gradient information," *Remote Sens.*, vol. 13, no. 12, p. 2382, Jun. 2021.
[30] J. Hu, M. Zhao, and Y. Li, "Hyperspectral image super-resolution by deep spatial–spectral exploitation," *Remote Sens.*, vol. 11, no. 10, p. 1229, May 2019.
[31] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3631–3640.
[32] O. Semerci, N. Hao, M. E. Kilmer, and E. L. Miller, "Tensor-based formulation and nuclear norm regularization for multienergy computed tomography," *IEEE Trans. Image Process.*, vol. 23, no. 4, pp. 1678–1693, Apr. 2014.
[33] T. Akgun, Y. Altunbasak, and R. M. Mersereau, "Super-resolution reconstruction of hyperspectral images," *IEEE Trans. Image Process.*, vol. 14, no. 11, pp. 1860–1875, Nov. 2005.
[34] H. K. Aggarwal and A. Majumdar, "Hyperspectral image denoising using spatio-spectral total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 442–446, Feb. 2016.
[35] M. A. Veganzones, M. Simoes, G. Licciardi, N. Yokoya, J. M. Bioucas-Dias, and J. Chanussot, "Hyperspectral super-resolution of locally low rank images from complementary multisource data," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 274–288, Jan. 2016.
[36] R. Arablouei and F. de Hoog, "Hyperspectral image recovery via hybrid regularization," *IEEE Trans. Image Process.*, vol. 25, no. 12, pp. 5649–5663, Dec. 2016.
[37] A. Torralba and A. Oliva, "Statistics of natural image categories," *Netw., Comput. Neural Syst.*, vol. 14, no. 3, pp. 391–412, Jan. 2003.
[38] M. Sadeghi, M. Babaie-Zadeh, and C. Jutten, "Learning overcomplete dictionaries based on atom-by-atom updating," *IEEE Trans. Signal Process.*, vol. 62, no. 4, pp. 883–891, Dec. 2013.
[39] M. Elad, "Why simple shrinkage is still relevant for redundant representations?" *IEEE Trans. Inf. Theory*, vol. 52, no. 12, pp. 5559–5569, Dec. 2006.
[40] J. Yang and Y. Zhang, "Alternating direction algorithms for ℓ_1-problems in compressive sensing," *SIAM J. Sci. Comput.*, vol. 33, no. 1, pp. 250–278, 2011.
[41] S. Jae, *Two-Dimensional Signal and Image Processing*. Upper Saddle River, NJ, USA: Prentice-Hall, 1990.
[42] Z. Stan, "Markov random field models in computer vision," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 1994.
[43] D. L. Donoho and I. M. Johnstone, "Ideal spatial adaptation by wavelet shrinkage," *Biometrika*, vol. 81, no. 3, pp. 425–455, Sep. 1994.
[44] Q. Zhang, R. Aliaga-Rossel, and P. Choi, "Denoising of gamma-ray signals by interval-dependent thresholds of wavelet analysis," *Meas. Sci. Technol.*, vol. 17, no. 4, pp. 731–735, Apr. 2006.
[45] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.
[46] M. Graña and B. Ayerdi. *Hyperspectral Remote Sensing Scenes Collected*. Accessed: Feb. 17, 2012. [Online]. Available: https://www.ehu.eus/ccwintco/index.php
[47] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[48] R. H. Yuhas, A. F. H. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. 3rd Annu. JPL Airborne Geosci. Workshop*, vol. 1, 1992, pp. 1–3.
[49] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion?" in *Proc. 3rd Conf. Fusion Earth Data, Merging Point Measurements, Raster Maps and Remotely Sensed Images*, 2000, pp. 99–103.
[50] R. C. Gonzales, *Digital Image Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 2007.
[51] C. Zuo et al., "Rotation invariant similarity measure for non-local self-similarity based image denoising," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 1618–1622.



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